

From Grid Microphysics to the Radial Acceleration Relation: A Minimal Gradient-Coupled Excitation Model

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EFC Paper Series

March 2026

Abstract. A precise structural gap exists in Energy-Flow Cosmology: the covariant effective field theory (EFT) requires a gravitational enhancement function $\mu(g) = 1/(\exp(\sqrt{g/a_0}) - 1)$ that is formally identical to a Bose–Einstein occupation number, but this form cannot be derived from any classical local Lagrangian. We show that a minimal microscopic model—discrete degrees of freedom on a substrate (grid) with gradient-coupled excitation energy $E \propto \sqrt{|\nabla\Phi|}$ —reproduces the observed functional form without free parameters beyond the measured scale a_0 . The key physical input is that excitation energy couples to local gravitational strain (gradient of the potential) rather than to the potential itself or to the density. This identifies the missing bridge between the EFC macro-framework (covariant action, GW theorem, slip cancellation, stability) and the micro-interpretation (grid nodes, bosonic modes, thermodynamic occupation). We state the model, derive the result, discuss its physical content and limitations, and identify the conditions under which it could be falsified.

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1 The Problem

1.1 The identified gap

In a companion paper [2], a systematic 17-iteration programme established five structural results for a covariant scalar–tensor EFT with an entropy-like scalar field $\mathcal{S}$ modifying gravitational dynamics. All five results—gravitational wave speed $c_{\text{gw}} = c$, gravitational slip $\eta \approx 1$, solar-system compatibility, ghost/tachyon freedom, and the identification of the Radial Acceleration Relation (RAR) as a Bose–Einstein occupation number—hold independently of the specific response function $\mu(g)$.

However, the programme also established a critical negative result: the classical field equation from the action $-\beta|\partial\mathcal{S}|^2$ produces a gravitational correction that *increases* with acceleration, whereas the RAR requires a correction that *decreases*. A classical local Lagrangian with finitely many derivatives generically produces polynomial corrections, not exponential suppression.

The gap is therefore not a vague “incompleteness” but a specific, localised obstruction: no classical local action can produce the Bose–Einstein form.

1.2 What is needed

The functional form to be reproduced is:

$$\mu(g) = \frac{1}{\exp(\sqrt{g/a_0}) - 1}, \quad (1)$$

which is the Bose–Einstein distribution $\langle n \rangle = 1/(e^{E/k_B T} - 1)$ with the identification $E/k_B T = \sqrt{g/a_0}$. This requires two ingredients: (i) bosonic degrees of freedom with a thermal or statistical occupation, and (ii) an excitation energy that scales as $E \propto \sqrt{g}$ where $g = |\nabla\Phi|$ is the local gravitational acceleration.

2 The Minimal Model

2.1 Physical picture

We postulate a substrate (grid) with discrete degrees of freedom at each node. Each node supports bosonic excitation modes whose energy depends on the local gravitational environment. The grid is in thermal equilibrium at a characteristic temperature T_{grid} , and the mean occupation number at each node determines the local gravitational enhancement.

This picture is motivated by, but logically independent of, the broader EFC ontology (Grid $\otimes$ Energy $\cdot$ Clarity). The derivation below requires only three assumptions, stated precisely.

2.2 Three assumptions

1. **Bosonic statistics.** The excitation modes at each grid node obey Bose–Einstein statistics. The mean occupation number is

$$\langle n \rangle = \frac{1}{e^{E/k_B T_{\text{grid}}} - 1}. \quad (2)$$

This is the standard result for non-interacting bosons. It requires only that the modes be discrete and indistinguishable—properties natural for collective excitations of a regular substrate.

2. **Gradient-coupled excitation energy.** The excitation energy at a node located at position $\mathbf{x}$ in a gravitational field with potential $\Phi(\mathbf{x})$ is

$$E(\mathbf{x}) = \hbar\omega_0 \sqrt{\frac{|\nabla\Phi(\mathbf{x})|}{g_*}}, \quad (3)$$

where ω_0 is a characteristic frequency and g_* is a reference acceleration. The physical content is that the node energy responds to local *strain* (the gradient of the potential), not to the potential itself or to the local matter density.

3. **Single scale.** The combination $\hbar\omega_0/k_B T_{\text{grid}}$ and g_* yield a single observable scale:

$$a_0 \equiv g_* \left(\frac{k_B T_{\text{grid}}}{\hbar\omega_0} \right)^2 = 1.2 \times 10^{-10} \text{ m/s}^2. \quad (4)$$

This is the measured SPARC value [1].

No other assumptions are made. In particular, no specific grid topology, dimensionality, or dynamics is assumed.

2.3 Derivation

Substituting Eq. (3) into Eq. (2) and using $g = |\nabla\Phi|$:

$$\langle n \rangle = \frac{1}{\exp\left(\frac{\hbar\omega_0}{k_B T_{\text{grid}}} \sqrt{\frac{g}{g_*}}\right) - 1}. \quad (5)$$

By the definition of a_0 in Eq. (4):

$$\frac{\hbar\omega_0}{k_B T_{\text{grid}}} \sqrt{\frac{g}{g_*}} = \sqrt{\frac{g}{g_* (k_B T_{\text{grid}}/\hbar\omega_0)^2}} = \sqrt{\frac{g}{a_0}}. \quad (6)$$

Therefore:

$$\boxed{\langle n \rangle = \frac{1}{\exp(\sqrt{g/a_0}) - 1} = \mu(g)} \quad (7)$$

The gravitational enhancement $\mu(g)$ is the mean occupation number of gradient-coupled bosonic excitations on the grid.

3 Why $\sqrt{g}$ and Not g or Φ

3.1 Gradient coupling is physically natural

In an elastic medium, the energy stored in a deformation is proportional to the strain. For a discrete substrate in a gravitational field, the relevant strain is the difference in potential between neighbouring nodes, which for a smooth field is $|\nabla\Phi| \cdot l_g$ where l_g is the node spacing. The excitation energy of a mode localised at a node then scales with $\sqrt{|\nabla\Phi|}$ if the mode frequency is set by the square root of the restoring force per unit displacement—the standard result for harmonic oscillators in an elastic medium (analogous to phonon dispersion).

This is not unique to the EFC grid. Any discrete structure embedded in a gravitational field with harmonic restoring forces will have excitation energies that scale as the square root of the local strain.

3.2 Why alternatives fail

This was established in iteration v7 of the construction programme [2]:

- Coupling to density ρ : gives $m_{\text{eff}} \propto \sqrt{G\rho}$, which is not a function of g alone and produces the wrong functional form.
- Coupling to potential Φ : gives $m_{\text{eff}} \propto \sqrt{|\Phi|}$, which is r -dependent and not universal (depends on total enclosed mass, not local field gradient).
- Coupling to acceleration $g = |\nabla\Phi|$: gives $E \propto \sqrt{g}$, which produces the BE form with one universal scale.

Only gradient coupling is consistent with the observed universality of the RAR.

4 Lattice Derivation of $E \propto \sqrt{g}$

The gradient-coupling assumption can be derived, not merely postulated, from a minimal lattice model. The argument has three steps, each using only standard physics.

4.1 Setup

Consider two adjacent grid nodes i and j separated by lattice spacing l_g in a gravitational field with local acceleration $g = |\nabla\Phi|$. A bosonic mode is localised at node i . The mode can be displaced by a small amount ξ along the inter-node direction.

4.2 Restoring force from the gravitational gradient

The gravitational force on the mode at position $\mathbf{x}$ is $F = -\nabla\Phi$. When the mode is displaced by ξ from its equilibrium position at node i , the force it experiences changes by

$$\Delta F = -\nabla\Phi|_{x_i+\xi} + \nabla\Phi|_{x_i} \approx -(\nabla^2\Phi)\xi. \quad (8)$$

On the lattice, the discrete Laplacian gives $\nabla^2\Phi \sim \Delta\Phi_{ij}/l_g^2$, valid in the smooth-field limit where Φ varies slowly across the lattice spacing ($l_g \ll$ local radius of curvature of Φ). Using $\Delta\Phi_{ij} \approx l_g g$:

$$\Delta F \approx -\frac{g}{l_g}\xi. \quad (9)$$

This is Hooke's law $\Delta F = -k_{\text{eff}}\xi$ with effective spring constant

$$k_{\text{eff}} = \frac{g}{l_g}. \quad (10)$$

The physical content is clear: a steeper gravitational gradient produces a stiffer restoring force on a displaced mode. This is the same mechanism that determines phonon frequencies in a lattice with position-dependent bond stiffness.

4.3 Mode frequency and excitation energy

For a harmonic oscillator with spring constant k_{eff} and effective mass m_{eff} :

$$\omega = \sqrt{\frac{k_{\text{eff}}}{m_{\text{eff}}}} = \sqrt{\frac{g}{m_{\text{eff}}l_g}}. \quad (11)$$

Therefore:

$$E = \hbar\omega = \hbar\sqrt{\frac{g}{m_{\text{eff}}l_g}} \propto \sqrt{g}. \quad (12)$$

The functional form $E \propto \sqrt{g}$ follows from $\omega = \sqrt{k/m}$ and $k \propto g$. Given a harmonic response of lattice modes to a gravitational gradient, the $\sqrt{g}$ scaling is not a choice but a structural consequence of oscillator physics. It does not depend on the specific values of m_{eff} or l_g , which are absorbed into the single observable scale a_0 via Eq. (4).

4.4 Robustness

This scaling is generic for any system where the restoring force on a mode is set by the local gradient of an external potential. It appears in phonon dispersion in inhomogeneous elastic media, in the WKB approximation for a particle in a slowly varying potential (where turning-point frequency $\omega \propto \sqrt{|V'|}$), and in the Unruh effect where an accelerated observer detects thermal radiation at temperature $T = \hbar a / (2\pi c k_B) \propto \sqrt{a^2}$. The EFC grid is one instance of a broader class of systems with this property.

5 Consistency with the EFT Framework

5.1 The bridge

The covariant EFT [2] establishes the macro-framework:

$$G_{\text{eff}}(g) = G_N(1 + \mu(g)) , \quad (13)$$

with five structural properties that hold for any decreasing $\mu(g)$. The present note supplies the micro-model that produces the specific form of μ . Together, they form a complete chain:

$$\text{Grid DOF} \xrightarrow{\text{gradient coupling}} E \propto \sqrt{g} \xrightarrow{\text{BE statistics}} \mu(g) \xrightarrow{\text{EFT}} G_{\text{eff}} \xrightarrow{\text{Poisson}} \text{RAR} . \quad (14)$$

5.2 Connection to GRAV pipeline

The discrete Graph-AQUAL solver [4] places AQUAL nonlinearity on a cubic lattice and recovers Newton ($g \propto 1/r^2$) and MOND ($g \propto 1/r$) limits from one functional (KT1: pass). The emergent prefactor $C \approx 2.32$ (KT2: pass) represents discrete IR renormalisation—the lattice geometry modifies the effective a_0 .

However, KT3 (mass scaling: $\beta = 0.29$ vs expected 0.50) shows that the transition radius is dominated by the bulk Λ -scale rather than by $\sqrt{GM/a_0}$. This is the GRAV-sector manifestation of the same gap: the classical discrete operator does not produce the correct mass-dependent scaling.

The present micro-model suggests a resolution: if the excitation energy at each node is $E \propto \sqrt{g}$ (not set by the bulk scale), then the transition radius should follow from the condition $\langle n \rangle \sim 1$, i.e. $\sqrt{g_{\text{trans}}/a_0} \sim 1$, giving $g_{\text{trans}} \sim a_0$ and therefore $r_{\text{trans}} \sim \sqrt{GM/a_0}$ —precisely the MOND scaling that KT3 is missing.

This identifies KT3's failure as a consequence of using a classical operator (which imposes the bulk scale) rather than the statistical occupation (which imposes the local gradient scale). A quantum or statistical treatment of the graph modes should restore $\beta = 0.5$.

6 What This Model Does Not Do

This section is essential for intellectual honesty and reviewer credibility.

1. **No dynamics.** The model specifies excitation energies and occupation numbers but does not derive the grid Hamiltonian or the time evolution of the modes. It is a kinematic, not dynamic, model.
2. **No grid topology.** No specific lattice structure, dimensionality, or node spacing is assumed. The result depends only on the existence of discrete bosonic modes with gradient-coupled energies.
3. **No quantum field theory.** The occupation number (2) is used as an input, not derived from canonical quantisation of a field on the grid. A full derivation would require specifying the grid action and performing the path integral.
4. **No cosmological derivation.** The model operates in the static, weak-field limit. Extension to cosmological perturbation theory (time-dependent $\mu(a, k)$) requires additional work.
5. **No derivation of a_0 .** The single scale a_0 is measured, not predicted. A deeper theory would derive a_0 from the grid parameters ($\omega_0, T_{\text{grid}}, g_*$).

7 Falsifiability

The model makes one structural prediction that distinguishes it from a pure fitting exercise:

Proposition 1. *If the gravitational enhancement $\mu(g)$ arises from bosonic occupation of gradient-coupled modes, then:*

1. μ must have exact BE form—not Fermi–Dirac, not classical Boltzmann, not a power law. Any deviation from the BE form at the level of galaxy-survey precision ($\lesssim 5\%$) would falsify the model.
2. The coupling variable must be $g = |\nabla\Phi|$, not ρ , Φ , or any other local quantity. Observational evidence for density-dependent or potential-dependent μ would falsify the model.
3. The excitation energy must scale as $\sqrt{g}$, not linearly. If future derivations from a specific grid Hamiltonian yield $E \propto g^\alpha$ with $\alpha \neq 1/2$, the model is falsified.

Additionally, the prediction that KT3 should recover $\beta = 0.5$ when the Graph-AQUAL solver incorporates statistical occupation (rather than classical field values) provides a concrete numerical test within the existing GRAV pipeline.

8 Conclusion

The gap between the EFC macro-framework and its micro-interpretation has been reduced to a single physical statement: excitation energy couples to the square root of the local gravitational gradient. Given this one input, Bose–Einstein statistics produce the observed RAR with no additional free parameters.

The derivation is minimal—three assumptions, five lines of algebra. It does not constitute a full quantum theory of the grid, but it identifies the precise physical content that any such theory must reproduce: gradient-coupled harmonic excitations with bosonic statistics. This connects the two previously isolated clusters in the EFC knowledge graph (covariant EFT and discrete GRAV) through a single, falsifiable bridge.

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